Critical Thinking (BES-103) Revision Notes

Prepared for Exam Review

April 2025

These notes cover **Lectures 3, 4, 5, and 7** for your Critical Thinking exam. Focus on definitions, examples, and connections to critical thinking. Use computer science (CS) scenarios (e.g., coding, projects) to apply concepts. Review quickly and think how ideas link.

1 Lecture 3: Reflection and Faulty Ways of Thinking

Key Concepts

- **Reflection**: Deep thinking about experiences to learn/grow.
  - *Why?* Builds **self-awareness** (know thoughts), **self-improvement** (fix weaknesses), **self-control** (change behavior).
  - *How?* **Reflective writing**: Describe (what?), Analyze (why?), Action (next?).
  - *Do's*: Honest, personal, life-connected.
  - *Don'ts*: No summarizing or complaining without thought.
  - *CS Example*: Reflect on buggy code—why failed? Plan better debugging.
- **Faulty Thinking** (3 traps):
  - **Filtering**: See only negatives (e.g., "My code's awful" despite good parts).
  - **Overgeneralizing**: One error means always bad (e.g., "Failed CS quiz, I'm terrible at coding").
  - **Catastrophizing**: Expect worst (e.g., "Bug will ruin project and grade").
- **Clock vs. Weather Vane**:
  - **Clock**: Think independently, steady reasoning.
  - **Weather Vane**: Swayed by others.
  - *CS Example*: Clock tests new tool; weather vane follows crowd.

Why It Matters

Reflection improves critical thinking by learning from mistakes. Avoiding faulty thinking ensures clear decisions (e.g., picking algorithms logically).

Exam Tips

Define reflection, explain a faulty thinking trap with example, or describe reflection in CS (e.g., debugging).

2 Lecture 4: Parameters of Thinking Skills

Key Concepts

- **Assessing Thinking Skills**: Measuring problem-solving, creativity, decisions.
  - *Methods*: Tests (quizzes), performance (tasks), observation (watching work).
  - *CS Example*: Teacher observes debugging to assess logic.
- **Detecting Emotional Manipulation**: Tricks clouding thinking.
  - *Signs*: Guilt-tripping (e.g., "Do all coding or we fail"), bullying, gaslighting (doubt self), love bombing (fake praise).
  - *CS Example*: Spot guilt in team to stay fair.
- **Focusing Attention**: Stay on task, no distractions.
  - *CS Example*: Code without phone to catch errors.
- **Identifying Differences**: Spot what's unique.
  - *CS Example*: Compare website designs—clear vs. cluttered.
- **Recognizing Sequences**: See patterns/order.
  - *CS Example*: Notice loop output (2, 4, 6) to predict/fix bugs.
- **Categorizing**: Sort info into groups.
  - *CS Example*: Group tasks—design, code, test—for organization.

Why It Matters

Skills build critical thinking (focus, analyze, organize). Avoiding manipulation keeps reasoning clear (e.g., pick tools by evidence).

Exam Tips

Define a skill (e.g., categorizing), give CS example (e.g., sorting data), explain manipulation's impact.

3 Lecture 5: Critical Thinking Barriers

Key Concepts

- **Egocentrism**: Thinking only of self.
  - **Self-Interested Thinking**: Choose for own benefit (e.g., pick CS tool you like, not best).

- **Self-Serving Bias**: Credit success, blame others for failure (e.g., “Code worked ‘cause I’m great; failed ‘cause bad specs”).

- **Sociocentrism**: Group is always right.
  - **Group Bias**: Favor group (e.g., “Our app’s best” without checking).
  - **Conformism**: Follow crowd (e.g., use trendy CS framework).
- **Unwarranted Assumptions**: Believe without proof.
  - *CS Example*: Assume tool’s bad (“nobody uses it”), no test.
- **Relativistic Thinking**: Truth is opinion.
  - **Subjectivism**: Truth is what *you* think (e.g., “My code’s fine ‘cause I like it”).
  - **Cultural Relativism**: Truth by culture (e.g., app design “right” locally, confuses globally).
- **Wishful Thinking**: Believe what you want.
  - *CS Example*: “My app will go viral” without tests.

Why It Matters

Barriers skew thinking, leading to bad choices. Avoiding them ensures fair, evidence-based decisions in CS.

Exam Tips

Define barrier (e.g., sociocentrism), give example, explain impact (e.g., conformism in projects).

4 Lecture 7: Argument and Non-Argument

Key Concepts

- **Key Terms**:
  - **Position**: View (e.g., “App needs login”).
  - **Agreement**: Share view (e.g., “I agree”).
  - **Disagreement**: Oppose view (e.g., “No, it doesn’t”).
  - **Argument**: Reasons to persuade (e.g., “Login protects data”).
- **Argument vs. Disagreement**:
  - Arguments have reasons; disagreements don’t.
  - *CS Example*: “Code’s bad” (disagreement) vs. “Code’s bad, $O(n^2)$ when $O(n)$ possible” (argument).
- **Non-Arguments**:
  - **Descriptions**: Neutral facts.

    - *CS Example*: "App loads in 3 sec" (no persuasion).
  - **Explanations**: Why/how, no persuasion.
    - *CS Example*: "App crashed: null pointer" (no argument).
    - *Note*: "So we should…" makes it an argument.
- **Distinguishing Arguments**: Find reasons amid descriptions, intros.
  - *CS Example*: "App uses Python. It's fast 'cause Python optimizes loops"—argument is "fast" part.

Why It Matters

Critical thinking builds/spots arguments with evidence, not opinions. In CS, justify choices (e.g., database) logically.

Exam Tips

Identify argument vs. non-argument, explain why, give CS argument example.

5 Connections Between Lectures

- **Reflection (L3) + Barriers (L5)**: Reflect to catch barriers (e.g., "Am I egocentric blaming others for bugs?").
- **Skills (L4) + Arguments (L7)**: Categorizing organizes arguments (e.g., reasons for tool choice). Focusing spots arguments.
- **Faulty Thinking (L3) + Barriers (L5)**: Overgeneralizing like unwarranted assumptions—skip evidence. Reflection helps.
- **Skills (L4) + Barriers (L5)**: Identifying differences fights group bias. Avoiding manipulation counters sociocentrism.
- *CS Application*: Reflect to improve coding, use sequences to plan, avoid wishful thinking in tests, argue for tech choices.

6 Quick Review Checklist

- **Definitions**: Know reflection, faulty thinking (filtering, etc.), skills (categorizing, etc.), barriers (egocentrism, etc.), argument, disagreement, description, explanation.
- **Examples**: Have CS example per concept (e.g., reflection for debugging, argument for algorithm).
- **Connections**: Link ideas (e.g., reflection fights barriers).
- **Spotting**: Identify arguments vs. non-arguments.
- **Reflection**: How you've used critical thinking (e.g., projects).

7 Last-Minute Tips for Exam

- **Morning Plan**: Skim notes (10-15 mins). Focus on **bold** terms, CS examples.
- **Key Focus**: Why concepts matter for critical thinking (e.g., arguments need reasons).
- **Quick Practice**: Pick barrier, think CS example.
- **Exam Strategy**: Read carefully. Define clearly, use CS examples, connect ideas.